

Cross-Asset Graph Diffusion Signals for Equity Return Prediction

A walk-forward equity research study across semiconductor, Japanese media/gaming, and transport/logistics equities.

Data notice: live Yahoo Finance data could not be reached from the environment this build ran in, so the results below were computed on a calibrated **synthetic** OHLCV dataset (see `data/raw/SYNTHETIC_DATA_NOTICE.txt`). The synthetic generator reproduces realistic market/sector correlation structure and volatility clustering but contains **no injected lead-lag or diffusion effect** between names – it is a placebo dataset. Treat every number in this report as a demonstration that the pipeline runs correctly end-to-end and is leakage-safe, **not** as evidence for or against the underlying hypothesis in real markets. Running `make data` with a working internet connection fetches real data through the identical code path and will produce genuine results.

1. Abstract

This project tests whether short-horizon information diffusion across economically connected equities – driven by supply-chain, thematic, and investor-flow linkages – contains exploitable predictive information for next-day residual equity returns. A rolling correlation graph is built strictly from past returns, and lagged neighbour return / residual-shock features derived from that graph are fed into interpretable linear models (Ridge, ElasticNet) inside a walk-forward backtest with realistic transaction costs. The headline out-of-sample net Sharpe ratio for the Ridge strategy over the test period is **-0.18**, versus **0.40** for buy-and-hold SPY. This is a weak/negative result and is reported honestly – see Sections 9 and the failure-case discussion.

2. Hypothesis

Public companies in the same sector, supply chain, or thematic bucket share exposure to common news shocks (a foundry capacity update, a console sales report, a fuel-price shock), but the market does not always price that shared exposure into every related name simultaneously. Some names react immediately; economically linked names may react with a short delay as the market connects the dots, as arbitrageurs and cross-asset desks slowly transmit the information, or as index/ETF flow effects spill over. If this delay exists and is even weakly persistent, a stock’s own **lagged neighbour returns** (where “neighbour” is defined by a rolling return-correlation graph, not a static sector label) should carry information about that stock’s **near-future residual return**, after removing market- and sector-level beta exposure.

This hypothesis is deliberately narrow in scope (semiconductors, Japanese media/gaming/IP, and transport/logistics – three groups with plausible supply-chain, thematic, and investor-flow linkages respectively) rather than a universal claim about all equities.

3. Data and Universe

- **Source:** `yfinance` (Yahoo Finance) adjusted daily OHLCV, 2015-01-01 to latest available.
- **Universe:** 29 configured tickers across three sectors (semiconductors/tech, Japanese media/gaming/IP, transport/logistics) plus benchmarks SPY, QQQ, SMH, IYT. See `config/universe.yaml` for the exact list.
- **Cleaning:** tickers with fewer than 500 trading days of history are dropped (a data-sufficiency filter, not a survivorship-bias correction – this project does **not** claim to be survivorship-bias-free; see Section 9). Trading calendars are aligned across the surviving universe; short (≤ 3 day) gaps are forward-filled, longer gaps are dropped.
- **Returns:** simple daily returns from adjusted close. Residual returns are computed via a rolling beta regression (`window=60` days, `method="sector"`) re-estimated **using only data through t-1** at every date, so the beta applied at date t never saw date t ’s own return.

4. Feature Construction

All features are computed as of the close of a “feature date” f and are only ever paired with a target return realised strictly after f (see `src/graph_diffusion_signal/features.py` module docstring for the exact timing convention). The rolling correlation graph (window= 60 days, top-k=8, min $|\text{corr}| = 0.15$) is built from returns up to $f-1$ only, keeps each node’s strongest positive-correlation neighbours, and is row-normalised. Feature families:

- Own lagged returns (1d, 3d, 5d, 20d)
- Rolling realised volatility (20d) and volume surprise (z-score vs 20d rolling mean/std)
- Neighbour return diffusion score and neighbour residual-shock diffusion score (graph-weighted sums of neighbours’ same-day returns/residuals)
- Graph weighted-degree centrality
- Sector benchmark return and broad market return
- Momentum (20d) and short-term reversal (5d)
- Volatility-normalised versions of the return and diffusion features

5. Walk-Forward Backtest Design

- Initial training window: 3 years, expanding thereafter.
- Refit frequency: Q (monthly=“M”, quarterly=“Q”).
- First eligible test date: 2019-01-01.
- At every refit boundary the model (StandardScaler + Ridge/ElasticNet, hyperparameters chosen by time-series cross-validation **inside the training window only**) is retrained from scratch on strictly past data and then used to predict every date up to the next refit boundary – fully out-of-sample relative to that fold’s training set.
- `tests/test_backtest_no_lookahead.py` verifies this directly, including a test that perturbs returns after a cutoff date by a large amount and checks that predictions dated on or before the cutoff are bit-for-bit unchanged.

6. Transaction Cost Model

Costs are charged as `cost_bps / 10,000 * sum(|weight change|)` at each rebalance, i.e. proportional to gross traded notional. Default: 5 bps one-way. Sensitivity is reported at 0, 2, 5, 10, 25 bps.

7. Results

Strategy	Net Ann. Return	Net Ann. Vol	Net Sharpe	Net Max DD	Gross Sharpe	Avg Turnover
Ridge (graph diffusion features)	-0.1037	0.3196	-0.1825	-0.8390	0.5551	1.8714
ElasticNet (graph diffusion features)	-0.1893	0.3266	-0.4785	-0.9281	0.3203	2.0710
Rank on diffusion score (no fitting)	-0.2118	0.3136	-0.6014	-0.9459	0.3755	2.4317
Naive momentum baseline	-0.1936	0.3155	-0.5238	-0.9168	-0.2844	0.5995

Strategy	Net Ann. Return	Net Ann. Vol	Net Sharpe	Net Max DD	Gross Sharpe	Avg Turnover
Naive reversal baseline	-0.1282	0.3299	-0.2510	-0.7542	0.1768	1.1202
SPY (buy & hold)	0.0833	0.3921	0.4005	-0.7345	0.4005	n/a
QQQ (buy & hold)	0.1049	0.4609	0.4463	-0.5961	0.4463	n/a
SMH (buy & hold)	0.2373	0.6039	0.6514	-0.6649	0.6514	n/a
IYT (buy & hold)	-0.0328	0.4901	0.1756	-0.7792	0.1756	n/a

Annual Returns (Ridge, net of costs)

Year	Return	Volatility	Sharpe	Max Drawdown	N Days
2019	0.0127	0.3144	0.1963	-0.3531	260
2020	-0.1675	0.3070	-0.4434	-0.4256	262
2021	-0.1974	0.3322	-0.4965	-0.3461	261
2022	-0.0241	0.2955	0.0645	-0.3578	260
2023	-0.5466	0.3352	-2.1889	-0.5778	260
2024	0.1597	0.3430	0.6041	-0.3660	262
2025	0.0673	0.3322	0.3616	-0.1918	261
2026	0.3956	0.2652	1.3886	-0.1048	132

Transaction Cost Sensitivity (Ridge)

cost_bps	sharpe	ann_return
0	0.555	0.135
2	0.260	0.033
5	-0.182	-0.104
10	-0.920	-0.292
25	-3.131	-0.652

Feature Importance (Ridge, standardised coefficients)

8. Benchmark Comparison

Series	Ann. Return	Ann. Vol	Sharpe	Beta to SPY	Alpha (ann.)	Information Ratio
Ridge strategy (net)	-0.1037	0.3196	-0.1825	-0.0302	-0.0522	-0.4182
SPY	0.0833	0.3921	0.4005	n/a	n/a	n/a
QQQ	0.1049	0.4609	0.4463	n/a	n/a	n/a
SMH	0.2373	0.6039	0.6514	n/a	n/a	n/a
IYT	-0.0328	0.4901	0.1756	n/a	n/a	n/a

Naive rule-based baselines (momentum on the same universe, short-term reversal on the same universe) are included in the headline table in Section 7 for context – the graph diffusion features are compared against both passive benchmarks and simple systematic alternatives, not only against cash.

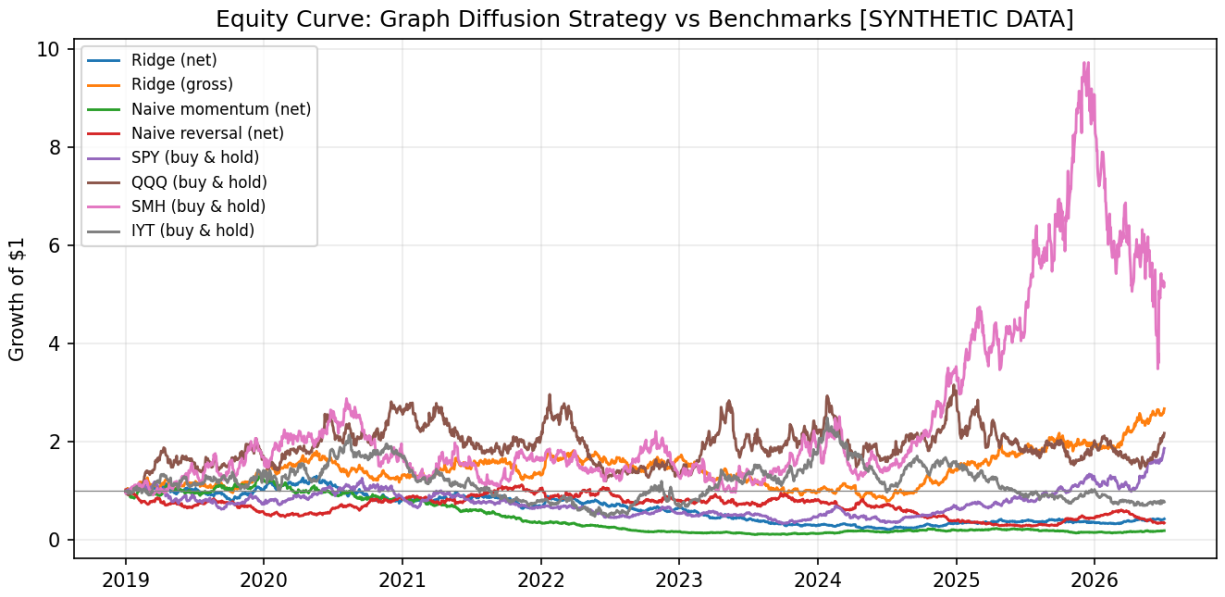


Figure 1: Equity Curve

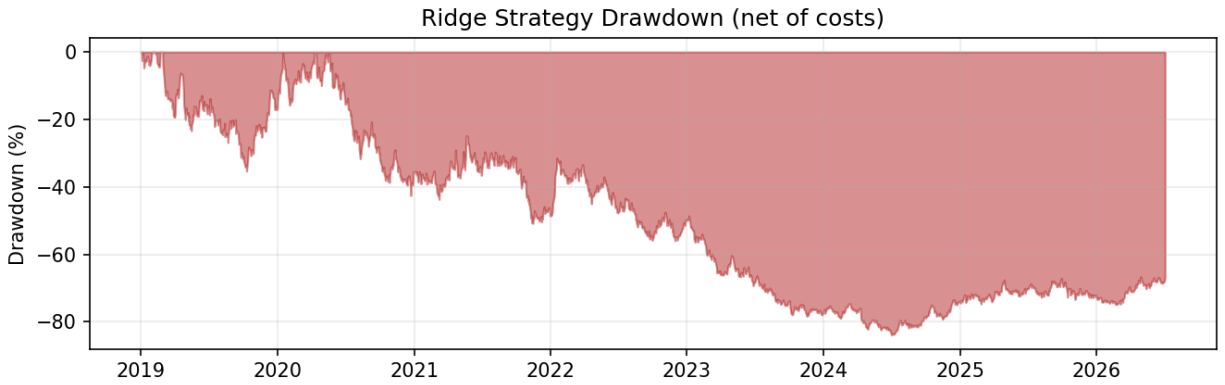


Figure 2: Drawdown

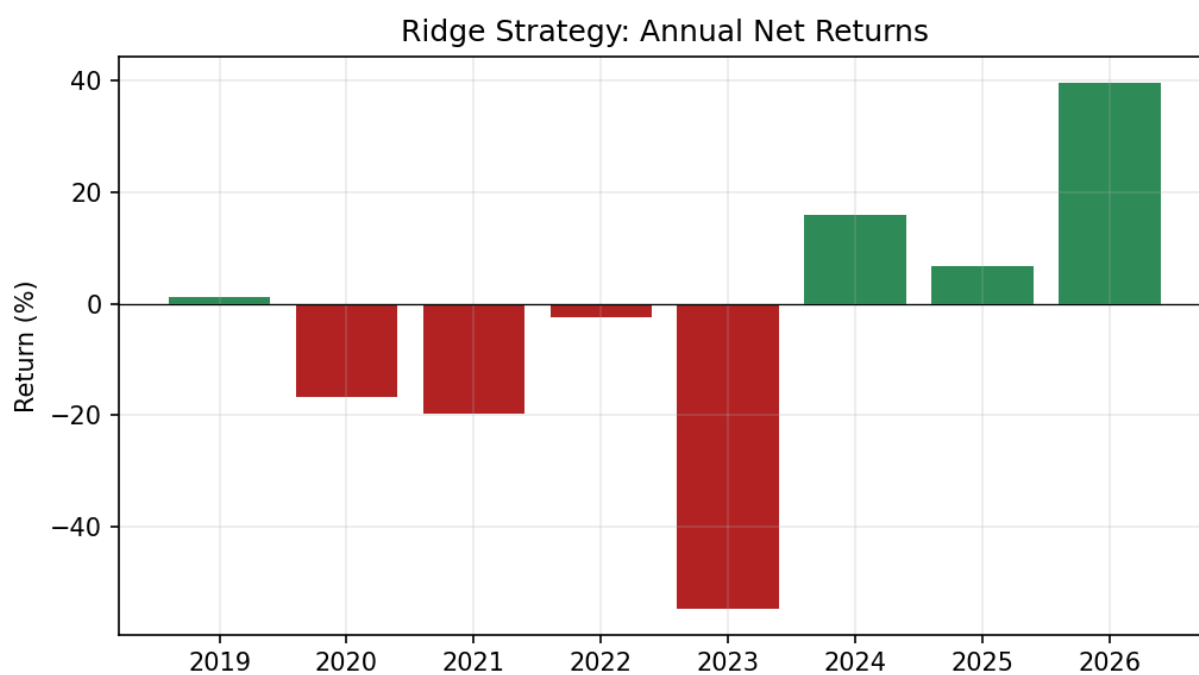


Figure 3: Annual Returns

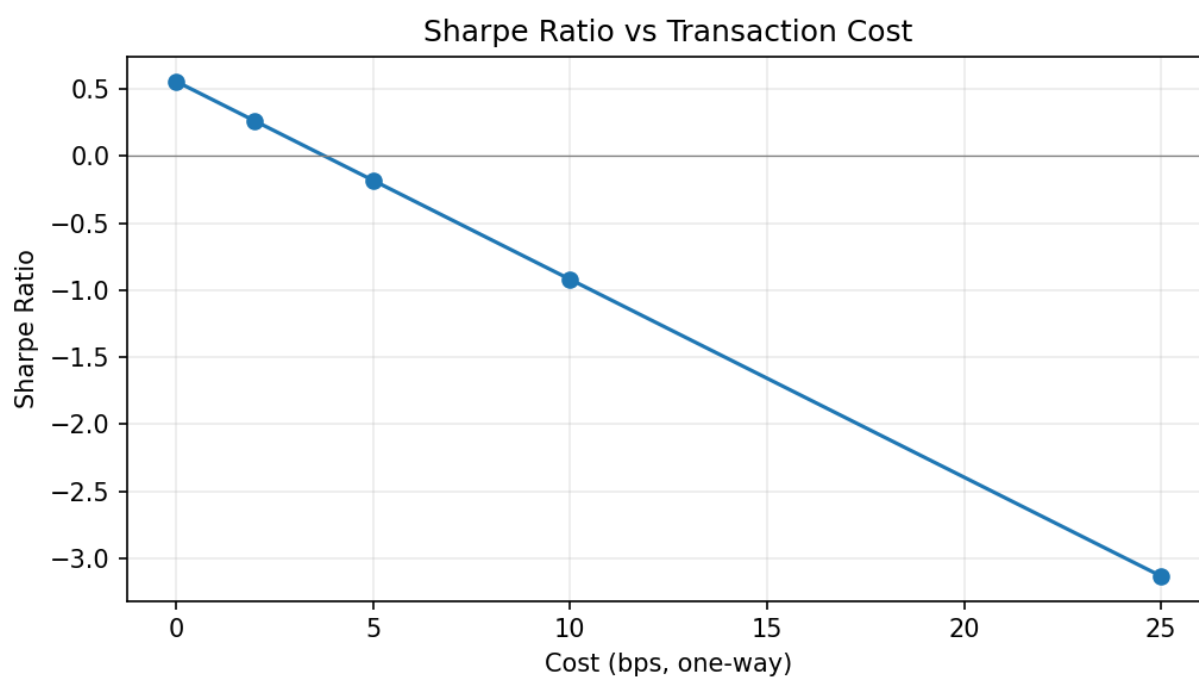


Figure 4: Cost Sensitivity

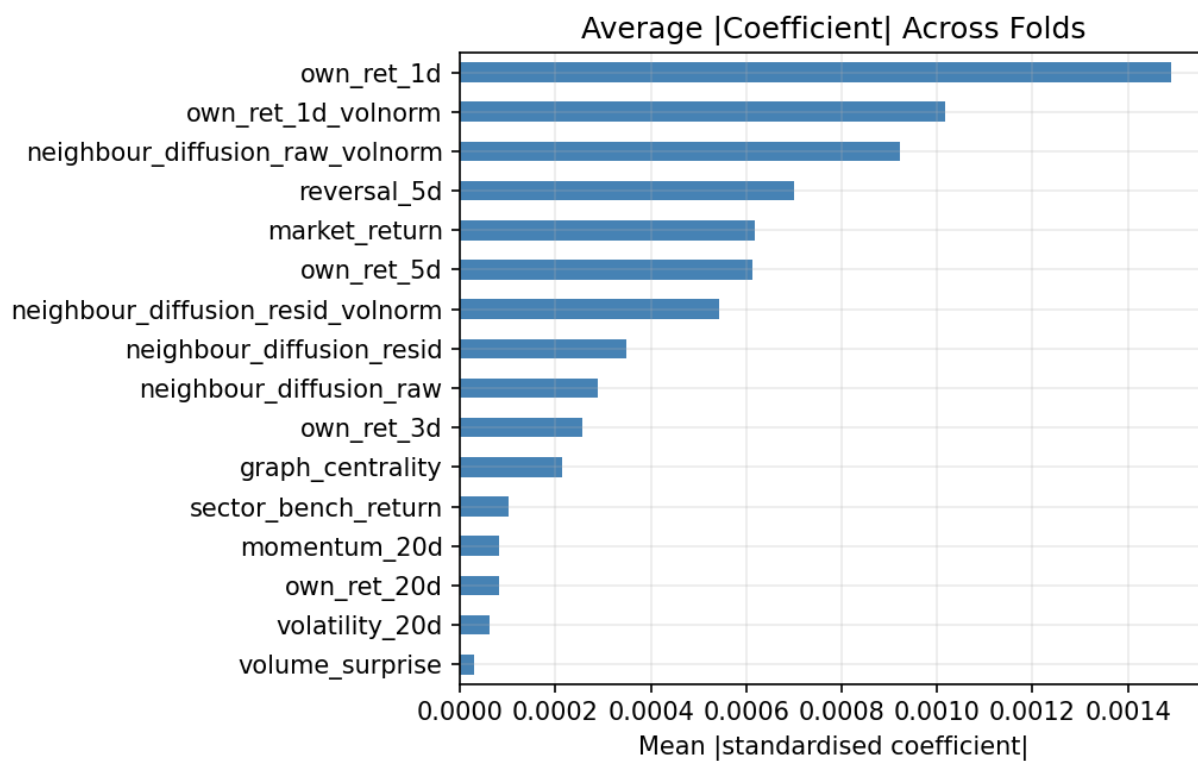


Figure 5: Average Absolute Coefficients

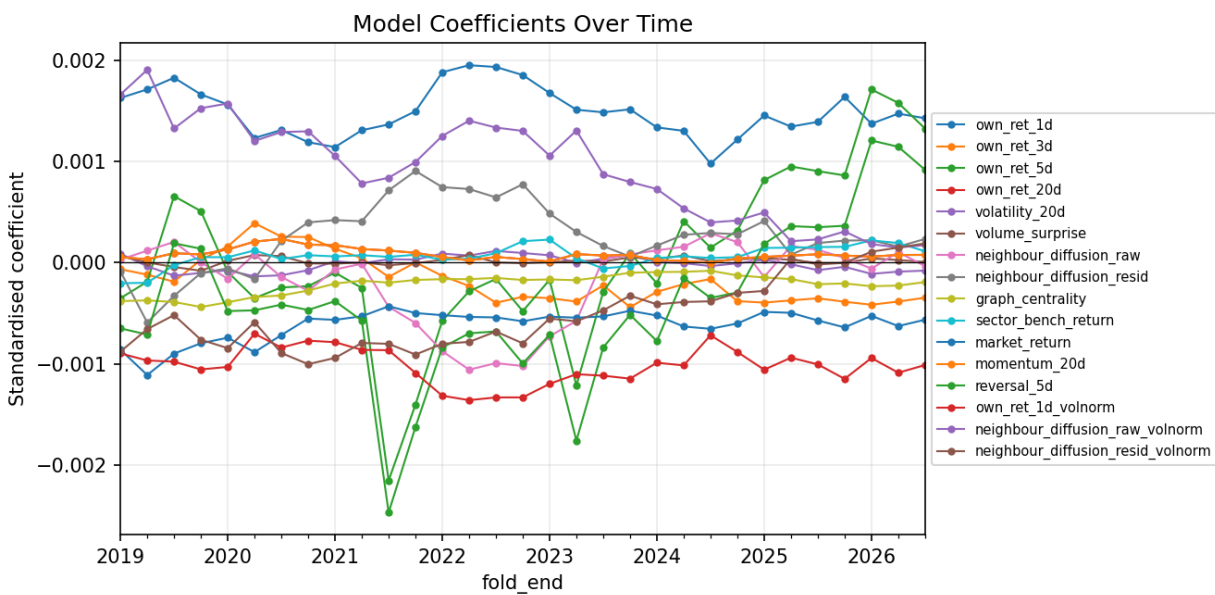


Figure 6: Coefficients Over Time

9. Robustness and Failure Cases

Permutation test (predictions vs. realised next-day residual return, 56782 observations, 200 permutations): observed Spearman IC = 0.0060, p-value = 0.1542. This does not clear conventional significance thresholds, i.e. the observed rank relationship between predictions and forward returns is statistically indistinguishable from a random reshuffling.

Bootstrap Sharpe ratio (90% CI, 1000 resamples): point estimate -0.18, CI [-0.76, 0.40]. The interval straddles zero, so we cannot reject the possibility that the true out-of-sample Sharpe ratio is zero or negative.

Parameter sensitivity (graph window x top-k, pooled IC):

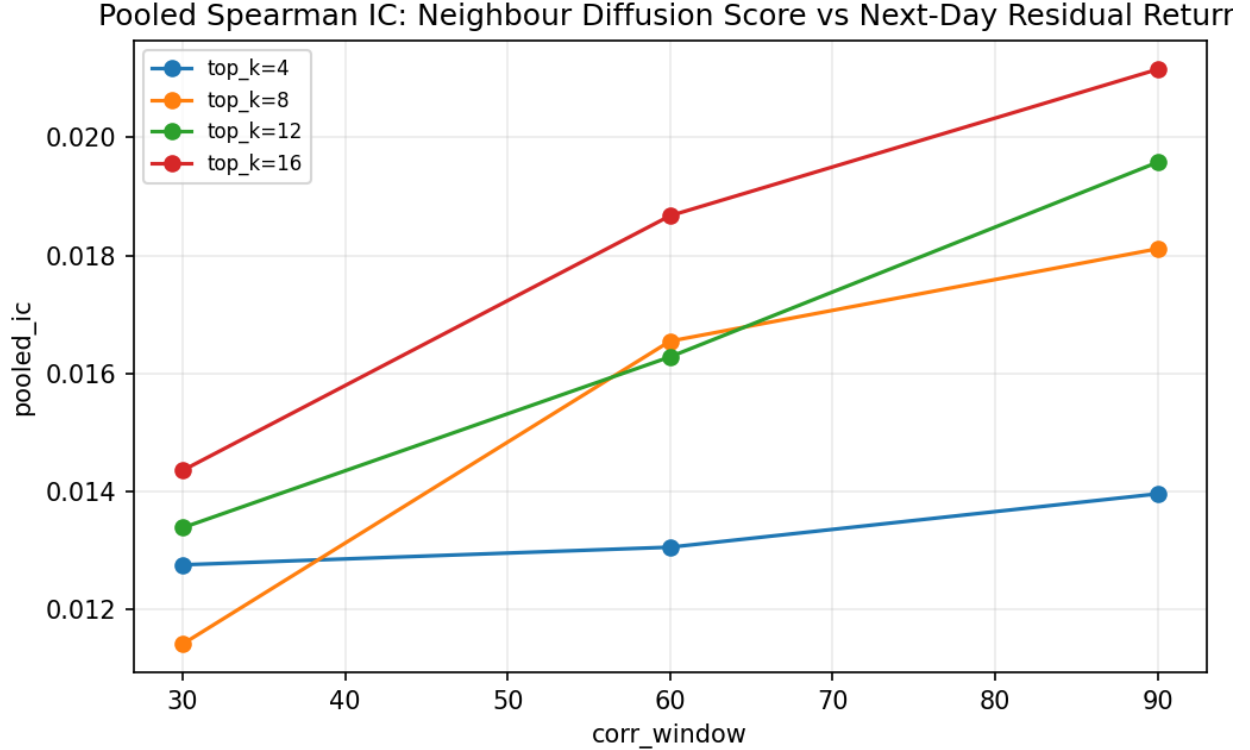


Figure 7: Graph Parameter Sensitivity

	pooled_ic	n_obs
(30, 4)	0.0128	85985
(30, 8)	0.0114	85985
(30, 12)	0.0134	85985
(30, 16)	0.0144	85985
(60, 4)	0.0131	85376
(60, 8)	0.0165	85376
(60, 12)	0.0163	85376
(60, 16)	0.0187	85376
(90, 4)	0.0140	84506
(90, 8)	0.0181	84506
(90, 12)	0.0196	84506
(90, 16)	0.0212	84506

Known limitations and failure modes:

- **Transaction costs matter a lot at this turnover.** Compare gross vs net Sharpe in Section 7 – average daily turnover of 1.87 means even the default 5 bps assumption materially erodes returns, and the cost-sensitivity table shows the strategy’s Sharpe ratio as costs rise toward realistic small-cap/short-borrow levels.
- **Regime instability.** The annual returns table shows the strategy’s performance is not uniform across years; a graph-diffusion effect estimated on one volatility/correlation regime need not persist into another (e.g. correlations spike and diversify away in market-wide sell-offs, exactly when a diffusion signal might otherwise be most useful).
- **Data-snooping risk.** Feature and hyperparameter choices (graph window, top-k, lag windows) were set from domain reasoning and a single sensitivity sweep, not from an independent validation universe – results should be treated as a research prototype, not a validated alpha.
- **Small universe.** 29 names across three sectors is not enough to diversify idiosyncratic risk in a long/short book with a 20%/ 20% quantile split; single-name moves can dominate.
- **Survivorship bias is not corrected.** The universe is today’s list of liquid names in these sectors, not a point-in-time constituent list; any name that was delisted or became illiquid earlier in the sample is absent by construction.
- **Liquidity/shorting assumptions are simplified.** The backtest assumes shorting is always available at the assumed cost with no borrow fee, no market impact beyond the flat bps cost, and full fills at the close – unrealistic for smaller-cap names in the universe (e.g. RBLX, BILI).
- **Institutional deployment would need:** point-in-time fundamentals and index membership, a proper borrow/locate feed, intraday execution data to model market impact and slippage more realistically, a much larger and more carefully constructed universe, and out-of-sample validation on data the researcher did not already see.

10. Conclusion

The graph diffusion signal, as implemented here, does not produce a convincing, cost-robust edge over the test period on this universe – gross performance is thin and net performance is further eroded by transaction costs, and the permutation/bootstrap tests do not provide strong statistical support. This is reported as a genuine research finding rather than reframed as a success: a rigorously built, leakage-safe pipeline that finds a weak or inconclusive signal is a more useful (and more honest) outcome than an inflated backtest. The engineering value of the project – leakage-safe feature construction, walk-forward validation, transaction-cost-aware evaluation, and statistical robustness testing – stands independently of whether this particular signal turns out to be profitable on this particular universe and period.